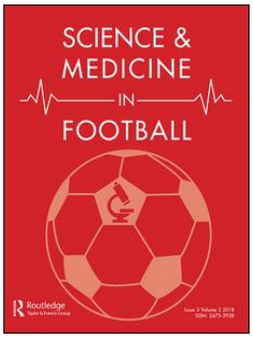

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Franco M. Impellizzeri , Martino V. Franchi , Fabio Sarto , Tim Meyer & Aaron J. Coutts

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EDITORIAL

Sharing information is probably more helpful than providing generic training recommendations on return to play after COVID-19 home confinement

Some authors of this editorial have recently published a call for awareness about the health consequences (e.g. increased injury risk) of home confinement in sport programming (Sarto et al. 2020). The main message of that editorial can be summarised as follows:

- We know that training restrictions and constraints may cause physiological decline.
- The increased risks are likely to be higher if the time needed to counteract this deconditioning is not granted before returning to competition. This may especially be the case for individuals whose level of training during home confinement has been constrained.
- Therefore, in deciding how and when to return to competition, it is important that sport organizations are aware that their decisions can have health consequences such as increased injury risk.

With most of us either directly or indirectly deriving benefit from the football industry, we are aware of the substantial financial and social pressures for resuming competitions. In the current environment, any decision regarding the recommencement of competition is difficult as there are numerous risks (i.e. financial, social and health) that must be considered and balanced (Carmody et al. 2020; Corsini et al. 2020; Duarte Muñoz and Meyer 2020). If the key decision makers are fully informed about the risks of these decisions, and all key stakeholders are willing to share the responsibility, this is perfectly fine. As stated in a famous quotation (wrongly) attributed to the Florentine political theorist Niccolò Macchiavelli: 'The ends justify the means'.

However, in the aforementioned editorial we deliberately chose not to provide recommendations for best approaches to return to competition. In contrast, there have been some articles providing suggestions on how to return to training and play after the COVID19 home confinement (Eirale et al. 2020; Jukic et al. 2020; Mohr et al. 2020). Although we commend others who have tried to provide recommendations for the return to competition, in this editorial we explain why we believe that providing specific recommendations in this situation is too challenging and generic advice may be not completely helpful.

Heterogeneity and unpredictable situations

Unfortunately, too many unpredictable individual and contextual factors make it difficult to provide recommendations that can be generalised to all situations. Indeed, even excluding the potential medical consequences of previous COVID-19 contagions in some athletes (Dores and Cardim 2020; Schellhorn et al. 2020), there is a high variability in the training and lifestyle behaviours employed during and immediately after the lockdowns. The training activities performed during home confinement are very heterogeneous between sports (e.g. different football codes), athletes, and sometimes even for players within the same team. This is due, for example, to: i) different fitness equipment and training facilities available to athletes during the lockdown, ii) country (different timing and restrictions of the lockdown), iii) psychological attitude, iv) competitive level and v) available support. Similarly, the return to training after the lockdown is dictated by the restrictions and norms adopted for reducing the risk of infections in the training environment, and these also differ between countries. In addition, the time allowed before the recommencement of official matches and changes in the competition calendar (and how congested it may be) is not only different between leagues and football codes, but in some countries is still yet to be defined. Moreover, an individualised approach is required, which is even more challenging in team sports. It is our position that providing 'general recommendations', which consists of generic training advice that resembles well established 'usual' training principles, results in little benefit to experienced practitioners. Indeed, coaches and support staff (medical, technical and scientific) are already aware and are already applying such training principles in their daily practice. Similarly, to our knowledge, individualised training approaches underlined by comparisons with reference data (e.g. historical physical tests and training load figures) are already adopted in many high-performance settings.

The challenge of providing evidence-based recommendations

It is conceivable that sport scientists, having a higher scientific knowledge and background on training issues, may be able to provide more 'evidence-based' recommendations. And this would normally be well accepted. Unfortunately, this seems difficult to achieve, as the available literature is mostly unrelated to relevant contexts (i.e. comparable to this unprecedented situation). This is even more challenging considering that a 'relevant context' is difficult to define given the aforementioned exceptional and heterogeneous factors. We are not arguing that recommendations are not possible in principle, but these should address and 'predict' actual specific and not the generic circumstances that players will encounter. For example, how can generic recommendations be applied in situations where the players are going to resume competitions within a limited (and variable) time available for

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reconditioning, even more so if the matches will be played every few days? The question here would be how to counteract athlete deconditioning, not in general, but specifically in that particular situation. Furthermore, any recommendations should be compatible with the constraints imposed by the regulations in place for preventing the spread of the infections. However, we are living in an evolving situation characterised by continuous adjustments made by public health and governmental authorities. This makes a prediction of these situations challenging. It follows that any training recommendation cannot be too generic, otherwise they would just be restating what coaches already know and do, and therefore not very useful.

How can sport science contribute to this situation?

While we appreciate the efforts of sport scientists to offer support to practitioners by the provision of training recommendations, there is a substantial risk that some advice would be based on selected and decontextualized literature, and may just result in common sense-based information. So, what can sport science do for helping practitioners? One of the challenges emerging from this extraordinary situation is the difficulty in managing information published in scientific journals or shared as preprints, that reaches the public in few minutes directly or through (social) media (Merchant and Lurie 2020). This reflects the high speed at which information can now be delivered to end users. However, the COVID-19 emergency has also shown an unprecedented level of interaction and information sharing among scientists (a topic that has as well been discussed in major medical journals such as Nature, Science, the Lancet and JAMA).

A similar role could be provided by sport scientists embedded in industry and the many academics who have strong collaborative links to sport organizations and teams. These individuals have the opportunity to network with practitioners, collect data and share information that can be used to better inform the community about the coalface experiences (with data) of teams and athletes during COVID-19. By favouring the circulation of this information, for example prioritising the publications on sport science journals (this is the intention of *Science and Medicine in Football*) and preprints, it would be possible to build a body of experiences that can inform any future decisions in similar circumstances (although we hope we do not have any further global lockdowns!). Clearly, the limitations of the observational studies need to be considered and we should be careful not to over-interpret them (as has happened with several COVID-19 related studies). However, this approach could provide important information on issues such as the feasibility (and effectiveness) of adapted training regimes and strategies, implementation issues, better characterization of the challenges that practitioners are confronted with and detailed descriptions of the training loads, injury incidence and typology, etc. While we acknowledge this as not being 'rocket science', such studies would serve to provide practitioners with more useful information than generic training advice. Finally, this same information could also be used to develop more specific training recommendations.

Disclosure statement

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Franco M. Impellizzeri
Human Performance Research Centre, Faculty of Health, University of Technology Sydney, Sydney, Australia
Franco.Impellizzeri@uts.edu.au

Martino V. Franchi and Fabio Sarto
Institute of Physiology, Department of Biomedical Sciences, University of Padua, Padua, Italy

Tim Meyer
Institute of Sports and Preventive Medicine, Saarland University, Saarbrücken, Germany

Aaron J. Coutts
Human Performance Research Centre, Faculty of Health, University of Technology Sydney, Sydney, Australia